

Credit Risk Prediction

Quick project summary for review

Objective

Estimate whether a loan may default using applicant, loan, and credit-history features. The current classifier is a Random Forest model; the web-app architecture lets a user submit applicant details and receive a predicted class and estimated default probability.

Data and workflow

Source workbook: **Credit risk dataset.xlsx**. It contains 32,581 rows and 12 columns. Removing 165 exact duplicate rows leaves 32,416 rows. The target is **loan_status** (0 = non-default, 1 = default in this dataset context). Missing values were present in **person_emp_length** (895) and **loan_int_rate** (3,116) before imputation.

Feature engineering includes monthly income, monthly interest burden, credit-history-to-age ratio, monthly interest burden relative to monthly income, and income minus loan amount. The modeling workflow uses a stratified 80/20 train-test split and categorical encoding.

Holdout test results

Metric	Result	Plain-English meaning
Accuracy	93.35%	Overall share of correct classifications
Precision (default class)	96.25%	Of loans predicted as default, share that were defaults
Recall (default class)	72.43%	Of actual defaults, share detected by the model
F1-score (default class)	82.66%	Balance of default precision and recall
ROC-AUC	93.31%	Ability to rank default cases above non-default cases

Confusion matrix (actual rows × predicted columns)

	Predicted 0	Predicted 1
Actual 0	5,026	40
Actual 1	391	1,027

Conclusion

The model achieved 93.35% accuracy and a 0.933 ROC-AUC on this single holdout split. However, it detected 72.43% of actual defaults, missing 391 of 1,418 default cases. Accuracy alone therefore hides an important credit-risk trade-off. Before any real lending use, compare models, review calibration and threshold choices, validate on separate data, and assess fairness and business costs.

Scope note

This is a learning/demo project, not a validated credit decision system. Metrics were reproduced using the supplied dataset, duplicate removal, training-set median imputation, the described feature engineering, categorical encoding, and a `RandomForestClassifier(random_state=42)` with a stratified 80/20 split (`random_state=42`). Results may vary if the code, dependencies, data, or split change. Power BI is intentionally excluded from the updated architecture; Excel logging remains part of the planned application flow.